Network Security - Important Write-up Questions

Q1. Explain the fundamental principles of network security. Discuss the objectives of Confidentiality, Integrity, Availability (CIA Triad), Authentication, Authorization, and Accountability with suitable examples.

Q2. What is a Firewall? Explain the different types of firewalls in detail. Compare Packet Filtering Firewall, Stateful Inspection Firewall, Proxy Firewall, and Next-Generation Firewall (NGFW) with their advantages and limitations.

Q3. Explain Linux Firewall (iptables) architecture and working. Discuss tables, chains, targets, and commonly used commands with suitable examples.

Q4. Describe the architecture and working of an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS). Compare IDS and IPS, explain their types, deployment methods, advantages, and limitations.

Q5. Explain the complete intrusion handling process, including intrusion detection, investigation, verification, recovery, reporting, documentation, and incident response. Why is proper documentation important in cybersecurity?

Q6. Discuss different network attacks including IP Spoofing, DoS, and DDoS attacks. Explain their working principles, impact on organizations, and various detection and mitigation techniques.

Q7. Explain the concept of Traffic Monitoring, Traffic Shaping, and Quality of Service (QoS). How do these techniques improve network performance and security? Discuss with practical examples.

Q8. What is a Virtual Private Network (VPN)? Explain different VPN types, VPN protocols, deployment methods, performance tuning techniques, troubleshooting methods, and common VPN security issues.

Q9. Explain the concept of DMZ (Demilitarized Zone), Virtual Hosts, Reverse Proxy, and Web Application Firewall (WAF). Discuss their architecture, working, advantages, and role in securing enterprise web applications.

Q10. Write detailed notes on Unified Threat Management (UTM), Threat Hunting Model, Packet Signature Analysis, and Security Policies. Explain how these technologies work together to provide a layered defense strategy in modern enterprise networks.